



INTERNSPARK

NETWORK PACKET ANALYSIS REPORT

Network traffic analysis identifying TCP handshake behavior, encrypted HTTP/DNS communication patterns, retransmissions, and abnormal local connection failures through packet capture inspection by using wireshark

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Introduction:

Objective:

The objective of this assessment was to capture and analyze network packets in order to identify communication protocols, TCP connection behavior, and suspicious or unusual network activity.

Scope:

The analysis focused on:

- HTTP/HTTPS traffic
- DNS communication
- TCP handshake behavior
- Packet flags and structures
- Identification of unusual or malformed packets

Tools Used:

Wireshark

Used for:

- Capturing network traffic
- Analyzing packet structures
- Monitoring protocol behavior
- Detecting abnormal traffic patterns

Methodology:

The packet analysis assessment was performed using a structured network monitoring approach. Traffic generated by the local system was captured and analyzed in real time using Wireshark.

The methodology focused on identifying:

- TCP handshake behavior
- HTTP/HTTPS communication
- DNS resolution activity
- Packet structures and flags
- Suspicious or abnormal traffic patterns

Steps involved:

Step 1 — Packet Capture:

Network traffic was captured while performing normal browsing and local application activity. Packet captures included:

- TCP traffic
- HTTPS communication
- DNS queries
- QUIC protocol traffic

The capture process monitored both:

- Localhost communication
- External internet communication

Step 2 — Traffic Filtering:

Wireshark display filters were used to isolate specific packet types for analysis.

TCP Handshake Filter

```
"tcp.flags.syn == 1 && tcp.flags.ack == 0"
```

This filter displayed TCP SYN packets used to initiate network connections.

Step 3 — Protocol Analysis:

Captured packets were analyzed to identify:

- Active protocols
- Packet flow behavior
- Connection establishment sequences
- Encrypted communication sessions

Protocols identified during analysis included:

- TCP
- DNS
- TLSv1.2
- TLSv1.3
- QUIC (HTTP/3)

Step 4 — Packet Structure Inspection:

Individual packets were inspected to analyze:

- Source and destination addresses
- Port numbers
- TCP flags
- TLS handshake messages
- QUIC frame types

Special attention was given to:

- SYN packets
- ACK responses
- Reset packets (RST)
- Retransmissions
- Encrypted application data

Step 5 — Identification of Abnormal Traffic:

Captured traffic was examined for unusual behavior such as:

- Failed TCP handshakes
- Continuous retransmissions
- Port reuse warnings
- Reset connections
- Packet capture anomalies

Observed warnings included:

[TCP Retransmission]

[TCP Spurious Retransmission]

[TCP Port numbers reused]

[TCP Previous segment not captured]

Step 6 — Risk Evaluation:

The identified behaviors were analyzed to determine:

- Whether traffic was normal or suspicious
- Possible causes of failures
- Potential network or application issues
- Security impact and risk level

Traffic behavior was categorized as:

- Normal encrypted communication
- Local service failure
- Network instability
- Capture-related anomalies

TCP Handshake Analysis:

Capture Filter Used:

“tcp.flags.syn == 1 && tcp.flags.ack == 0”

Observations:

The packet capture identified multiple TCP SYN packets originating from:

- 127.0.0.1
- ::1

Destination ports observed:

- 1521 (Oracle Database)
- 8080 (Web Server)
- 3000 (Development Server)

Behaviors Identified:

- Some SYN packets received no response
- Some connections received immediate: RST, ACK

responses indicating failed connections.

HTTP / HTTPS Traffic Analysis:

Ports Observed:

- 443 (HTTPS)

Protocols Identified:

- TLSv1.2
- TLSv1.3
- QUIC (HTTP/3)

TLS Handshake Observations:

Successful encrypted communication included:

- Client Hello
- Server Hello
- Encrypted Application Data

Server Name Indication (SNI) Detected:

- www.google.com
- fonts.gstatic.com
- www.gstatic.com

This confirms successful secure communication with external servers.

DNS Traffic Analysis:

DNS queries were observed before external encrypted communication.

Domain Resolution Activity:

DNS lookups were identified for:

- Google-related services
- External HTTPS resources

These resolutions were followed by TLS and QUIC communication sessions.

Packet Structure and Flag Analysis:

TCP Flags Observed:

Flag	Meaning	Observed Behavior
SYN	Start connection	Frequently observed
ACK	Acknowledgment	Present in responses
RST	Reset connection	Seen in failed local connections
PSH	Push data	Seen during HTTPS traffic
FIN	Close connection	Seen during retransmissions

Packet Behavior Analysis:

- Normal TCP Handshake:
Successful external communication followed the standard TCP handshake process:

SYN → SYN-ACK → ACK

This indicates healthy network communication.

- Failed TCP Handshake:
Some local services displayed failed connection attempts:

SYN → RST, ACK

This indicates that no active service was listening on the destination port.

- TCP Retransmissions:
The following retransmission events were observed:
 - [TCP Retransmission]
 - [TCP Spurious Retransmission]These indicate repeated packet transmission attempts.

QUIC Packet Structure:

QUIC traffic was identified as UDP-based encrypted communication.

QUIC Frame Types Observed:

- CRYPTO
- PING
- PADDING

TLS Packet Structure:

TLS traffic included:

- Client Hello
- Server Hello
- Change Cipher Spec
- Encrypted Application Data

This confirms successful encrypted sessions.

Unusual or Malformed Packet Analysis:

Issue 1: Broken TCP Handshakes:

Observation:
Repeated SYN packets were observed without successful handshake completion.
Immediate responses included: RST, ACK

Analysis:
This behavior indicates:

- Closed ports
- Stopped services
- No active listeners on:
- Port 8080
- Port 1521

Risk Level:
Medium

Issue 2: TCP Port Reuse:

Observation:
Wireshark identified:
[TCP Port numbers reused]

Analysis:
This indicates:

- Rapid retry attempts
- Repeated failed connections
- Possible application retry loops

Risk Level:
Medium

Issue 3: TCP Retransmissions:

Observation:
Normal and spurious retransmissions were detected.

Analysis:
Possible causes include:

- Packet loss
- Temporary network delay
- Wi-Fi instability
- Congestion

Risk Level:
Low

Additional Warnings Observed during analysis:

Warnings:

[TCP Previous segment not captured]
[Ignored Unknown Record]

Analysis:
These warnings are generally related to:

- Packet capture limitations
- Missing captured packets
- Non-malicious traffic anomalies

No malicious activity was identified.

Executive Summary:

The packet capture analysis identified two major network behaviors:

1. Failed local TCP connections to ports 8080 and 1521
2. Successful secure external communication using TLS and QUIC protocols

Technical Findings:

Local Service Failures:
Continuous SYN packets followed by reset responses indicate:

- Services not running
- Misconfigured applications
- Closed listening ports

Application Retry Behavior:
Rapid retries and TCP port reuse suggest:

- Poor retry handling
- Possible infinite connection retry loops

Healthy External Traffic:
Successful TLS and QUIC handshakes confirm:

- Proper encrypted communication
- Secure internet connectivity

Network Stability:
Minor retransmissions indicate:

- Temporary packet loss
- Normal network delays

No evidence of malicious traffic was observed.

Risk Classification:

Issue	Risk Level
Local service failure	Medium
TCP retransmissions	Low
External encrypted communication	No Risk

Recommendations:

Service Verification:

Start and verify required services:

- Oracle Database (1521)
- Web Server (8080)

Port Configuration:

Verify:

- Listening ports
- Service bindings
- Firewall rules

Application Improvements:

- Fix retry logic
- Prevent continuous reconnection attempts
- Implement proper timeout handling

Network Monitoring:

- Monitor retransmission frequency
- Check network stability if retransmissions increase

Conclusion:

The packet analysis revealed failed local service connections caused by unavailable or closed ports, along with repeated retry attempts generated by local applications. Despite local service failures, external internet communication was functioning securely using TLS and QUIC protocols.

Overall, the environment demonstrated:

- Secure encrypted external communication
- Minor network retransmissions
- Local service configuration issues requiring remediation

Screenshots:

No	Time	Source	Destination	Protocol	Length	Info
208	0.074554300	2409:4091:7:13::F5.	2404::6800:4800:806::	QUIC	1292	Initial, SCDID=96354cd6466106b, PKIN: 3, CRYPTO, P1
209	0.107507300	2409:4091:7:13::F5.	2404::6800:4800:816::	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
210	0.107507400	2409:4091:7:13::F5.	2404::6800:4800:816::	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
211	0.123482200	2409:4091:7:13::F5.	2404::6800:4800:825::	QUIC	1292	Initial, SCDID=214ab6d74ca1f7ff, PKIN: 4, PING, PING_ACK
330	0.143618600	2409:4091:7:13::F5.	2404::6800:4800:825::	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
339	0.149995100	2404::6800:4800:806::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=96354cd6466106b, PKIN: 3, CRYPTO, P1
340	0.152438100	2409:4091:7:13::F5.	2404::6800:4800:834::	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
383	0.166788800	2409:4091:7:13::F5.	2404::6800:4800:80F::	QUIC	1292	Initial, SCDID=b6585c8ed5d15aa6, PKIN: 1, PADDING, C
385	0.167717700	2409:4091:7:13::F5.	2404::6800:4800:80F::	QUIC	1292	Initial, SCDID=b6585c8ed5d15aa6, PKIN: 4, PING, CRYPTO
391	0.167568000	2409:4091:7:13::F5.	2404::6800:4800:80F::	QUIC	1292	Initial, SCDID=b6585c8ed5d15aa6, PKIN: 4, PING, CRYPTO
392	0.167565600	2404::6800:4800:825::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 4, CRYPTO, P1
400	0.196668000	2404::6800:4800:816::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
413	0.188112000	2409:4091:7:13::F5.	2404::6800:4800:816::	QUIC	1292	Initial, SCDID=96354cd6466106b, PKIN: 3, CRYPTO, P1
440	0.193513500	2404::6800:4800:825::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
463	0.213784000	2404::6800:4800:814::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=759e142dc6c73, PKIN: 3, CRYPTO, P1
482	0.214508000	2409:4091:7:13::F5.	2404::6800:4800:816::	QUIC	1292	Initial, SCDID=96354cd6466106b, PKIN: 3, CRYPTO, P1
484	0.230759100	2404::6800:4800:814::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1
494	0.952471200	2409:4091:7:13::F5.	2404::6800:4800:817::	QUIC	1292	Initial, SCDID=208724d4f0b80b3d, PKIN: 3, CRYPTO, C
1767	1.008007000	2409:4091:7:13::F5.	2404::6800:4800:817::	QUIC	1292	Initial, SCDID=208724d4f0b80b3d, PKIN: 3, CRYPTO, C
1783	1.262222600	2409:4091:7:13::F5.	2404::6800:4800:817::	QUIC	1292	Initial, SCDID=369e2d505a9574, PKIN: 2, PING, PING
1788	1.308889000	2409:4091:7:13::F5.	2404::6800:4800:817::	QUIC	1292	Initial, SCDID=369e2d505a9574, PKIN: 3, CRYPTO, P1
1801	1.457536000	2409:4091:7:13::F5.	2404::6800:4800:817::	QUIC	1292	Initial, SCDID=369e2d505a9574, PKIN: 3, CRYPTO, P1
1855	2.067707900	2606::4700:809f::6e6	2409:4091:7:13::F5.	TLSv1.2	134	Application Data
1922	2.439982400	2409:4091:7:13::F5.	2404::6800:4800:81d::	QUIC	1292	Initial, SCDID=89f717a9b30b1d, PKIN: 2, CRYPTO, P1
1926	2.535082000	2404::6800:4800:817::	2409:4091:7:13::F5.	QUIC	1292	Initial, SCDID=36c4a765943b0d1, PKIN: 3, CRYPTO, P1

Frame 2062: Packet, 1294 bytes on wire (10352 bits), 1294 bytes captured (10352 bits) on interface 0, 1294 bytes from 2409:4091:7:13::F5 (2409:4091:7:13::F5) to 2404::6800:4800:816:: (2404::6800:4800:816::), Seq: 240454560, Win: 0, Len: 1294

User Datagram Protocol, Src Port: 443, Dest Port: 55887

QUIC KEY

Packet (1294 bytes) Decrypted QUIC (1193 bytes) Reassembled TLS Handshake

tls filter result

No.	Source	Destination	Protocol	Length	Info
71	6.84.7685180	192.168.31.49	DNS	87	Standard query 653800 HTTPS www.gstatic.com
72	6.84.7684880	192.168.31.49	DNS	87	Standard query 6538F AAAA www.gstatic.com
76	6.8475000	192.168.31.49	DNS	87	Standard query 65fccc A www.gstatic.com
78	6.8476440	192.168.31.49	DNS	86	Standard query 653800 HTTPS www.gstatic.com
78	6.85150000	192.168.31.49	DNS	86	Standard query 6b9f84 AAAA www.google.com
80	6.8476000	192.168.31.49	DNS	86	Standard query 653800 HTTPS www.gstatic.com
102	6.8547300	192.168.31.49	DNS	146	Standard query response 653800 HTTPS www.gstatic.com
102	6.85473300	192.168.31.49	DNS	117	Standard query response 6538F AAAA www.gstatic.com
102	6.85473700	192.168.31.49	DNS	105	Standard query response 653800 HTTPS www.gstatic.com
118	6.85668700	192.168.31.49	DNS	165	Standard query response 65fccc A www.gstatic.com
129	6.85695900	192.168.31.49	DNS	113	Standard query response 65b8d HTTPS www.google.com
130	6.85696100	192.168.31.49	DNS	113	Standard query response 65b8d HTTPS www.google.com
135	6.87012600	192.168.31.49	DNS	97	Standard query 65a29 AAAA lh3.googleusercontent.com
137	6.87024500	192.168.31.49	DNS	97	Standard query 65b88 HTTPS lh3.googleusercontent.com
142	6.87026000	192.168.31.49	DNS	97	Standard query 65b88 HTTPS lh3.googleusercontent.com
162	6.87652900	192.168.31.49	DNS	95	Standard query 65dc5c HTTPS safesearch.google.com
164	6.87677700	192.168.31.49	DNS	95	Standard query 65385 AAAA safesearch.google.com
166	6.87677800	192.168.31.49	DNS	95	Standard query 65385 HTTPS safesearch.google.com
168	6.87745900	192.168.31.49	DNS	90	Standard query 65b967 A gengine.gstatic.com
170	6.87747500	192.168.31.49	DNS	90	Standard query 65b9f5 HTTPS gengine.gstatic.com
180	6.87751100	192.168.31.49	DNS	90	Standard query 653800 AAAA gengine.gstatic.com
186	6.89133900	192.168.31.49	DNS	185	Standard query response 65b88 HTTPS lh3.googleusercontent.com
186	6.89136200	192.168.31.49	DNS	154	Standard query response 65403 A lh3.googleusercontent.com
202	6.89136300	192.168.31.49	DNS	154	Standard query response 65403 A lh3.googleusercontent.com
212	6.89131300	192.168.31.49	DNS	96	Standard query 65591 HTTPS www.googletransparency.com

dns filter result

The image displays a Wireshark packet capture analysis of an HTTP GET request. The packet list on the left shows a sequence of packets, with packet 292 being the selected GET request. The packet details pane on the right shows the structure of the HTTP request, including the Host, User-Agent, and Accept headers. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
292	38.10837200	127.0.0.1	127.0.0.1	HTTP	736	GET / HTTP/1.1
296	38.11521200	127.0.0.1	127.0.0.1	HTTP	437	HTTP/1.1 304 Not Modified
330	38.21639700	127.0.0.1	127.0.0.1	HTTP	672	GET /chunk-24812242.js HTTP/1.1
331	38.21639700	127.0.0.1	127.0.0.1	HTTP	674	GET /chunk-24812242.js HTTP/1.1
346	38.22095300	127.0.0.1	127.0.0.1	HTTP	437	HTTP/1.1 304 Not Modified
350	38.22168400	127.0.0.1	127.0.0.1	HTTP	435	HTTP/1.1 304 Not Modified
352	38.22391700	127.0.0.1	127.0.0.1	HTTP	437	HTTP/1.1 304 Not Modified
354	38.22513800	127.0.0.1	127.0.0.1	HTTP	437	HTTP/1.1 304 Not Modified
361	38.23097500	127.0.0.1	127.0.0.1	HTTP	672	GET /chunk-7205845.js HTTP/1.1
363	38.23164200	127.0.0.1	127.0.0.1	HTTP	653	GET /polyfills.js HTTP/1.1
371	38.23260500	127.0.0.1	127.0.0.1	HTTP	654	GET /polyfills.js HTTP/1.1
374	38.23621800	127.0.0.1	127.0.0.1	HTTP	435	HTTP/1.1 304 Not Modified
378	38.23786900	127.0.0.1	127.0.0.1	HTTP	436	HTTP/1.1 304 Not Modified
381	38.23755100	127.0.0.1	127.0.0.1	HTTP	673	GET /chunk-3395245.js HTTP/1.1
389	38.23881700	127.0.0.1	127.0.0.1	HTTP	436	HTTP/1.1 304 Not Modified
393	38.23970500	127.0.0.1	127.0.0.1	HTTP	656	GET /main.js HTTP/1.1
405	38.24015800	127.0.0.1	127.0.0.1	HTTP	436	HTTP/1.1 304 Not Modified
411	38.24015800	127.0.0.1	127.0.0.1	HTTP	436	HTTP/1.1 304 Not Modified
420	38.24256900	127.0.0.1	127.0.0.1	HTTP	684	GET /styles.css HTTP/1.1
433	38.24841400	127.0.0.1	127.0.0.1	HTTP	437	HTTP/1.1 304 Not Modified
437	38.25001700	127.0.0.1	127.0.0.1	HTTP	597	GET /main.js?__proto__=Object&port=collingtest&Pts4eud HTTP/1.1
493	38.55858400	127.0.0.1	127.0.0.1	HTTP	370	HTTP/1.1 200 OK (text/plain)
501	38.60216700	127.0.0.1	127.0.0.1	HTTP	689	GET /assets/linen_jsn HTTP/1.1
503	38.60621500	127.0.0.1	127.0.0.1	HTTP	436	HTTP/1.1 304 Not Modified
537	38.72027200	127.0.0.1	127.0.0.1	HTTP	672	GET /test/admin/application-configuration HTTP/1.1

Frame 292: Packet, 736 bytes on wire (5888 bits), 736 bytes captured (5888 bits) on interface 0, 27.000000000 seconds on interface 0, 127.0.0.1

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 10458, Dst Port: 3000, Seq: 3810837200, Win: 0, Len: 0

Hypertext Transfer Protocol

Hypertext Transfer Protocol Protocol

Packets: 4422 - Displayed: 184 (4.2%)

Profile Details

http filter result

The image shows a Wireshark packet capture of a series of TCP connections. The packet list on the left shows the following details:

- 280 0.780727900 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 281 0.780818200 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 282 0.870184100 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 283 0.871054000 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 284 0.890691200 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 285 0.890695000 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 286 0.887534800 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 287 0.897612700 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 288 0.897769600 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 289 0.887788800 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 290 0.888082100 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 291 0.888165800 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 292 0.888377200 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 293 0.888444100 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 294 0.888979000 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 295 0.889308000 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 296 0.881523100 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 297 0.881523200 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 298 0.881199400 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 299 0.881195600 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 300 0.881394700 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 301 0.881392800 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 302 0.8814215800 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 303 0.8814313400 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0
- 304 0.8814378100 127.0.0.1 → 127.0.0.1 [ACK] Seq=18011408 Win=65428 Len=0

The packet details on the right show the following information:

- Frame 292: Packet, 736 bytes on wire (5888 bits), 736 bytes captured (5888 bits) on interface 0
- Ethernet II, Src: Intel(R) Ethernet Controller (P0-P7) 82:55:88:53:00:00, Dst: VirtualBox (NAT) 08:00:00:00:00:00
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Seq: 18011408, Len: 0
- Hypertext Transfer Protocol

The packet bytes on the right show the following data:

```
0000 02 00 00 00 00 00 02 c0 ba 20 00 00 00 00 3f 67
0010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0020 00 18 87 59 50 18 00 5f 56 3a 00 00 45 45 24
0030 2f 20 48 00 50 52 3f 21 3d 00 48 00 67 73 76
0040 20 60 6f 63 61 63 67 72 74 3a 33 00 00 00
```

tcp filter result

No.	Time	Source	Destination	Protocol	Length	Info
92	36.340418000	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
93	36.340422500	127.0.0.1	127.0.0.1	TCP	52	10141 → 8080 [ACK] Seq=1424143331 Len=0
108	36.603337900	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
109	36.853252500	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
110	36.853255000	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
115	37.105884400	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
117	37.109949200	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
143	37.359134000	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
144	37.362682800	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
183	37.614761300	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
184	37.614861300	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
243	37.867925200	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
245	37.867996700	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
250	38.119094600	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
280	38.318167900	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
285	38.370743200	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
436	38.378801400	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
490	38.520132000	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
495	38.559555800	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
587	38.622319300	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
589	38.622377400	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
593	38.632920000	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reuired: 10141 → 8080 [SVN] Seq=1424143331 Len=0]
510	38.639977800	127.0.0.1	127.0.0.1	TCP	44	8080 → 10141 [EST, ACK] Seq=1424143331 Len=0
685	38.854344400	127.0.0.1	127.0.0.1	TCP	56	10178 → 8080 [TCP Seq=1424143331 Len=0] Seq=1424143331 Len=0
690	38.854349000	127.0.0.1	127.0.0.1	TCP	44	8080 → 10178 [EST, ACK] Seq=1424143331 Len=0

```
tcp port == 8080 filter result
```

The image shows a Wireshark packet capture of a series of TCP connections. The packet list on the left shows 18 packets, all of which are TCP. The connections are established from 10.0.0.10 to 10.0.0.1. The packet details on the right show the structure of the TCP header, including the source and destination ports, sequence number, and flags. The packet bytes pane shows the raw data of the packets.

No.	Source	Destination	Protocol	Length	Info
0	6.0.0.0/20.0.0.0	127.0.0.1	TCP	44	40778 → 40777 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40777 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40778 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
14	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
16	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
18	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
20	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
22	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
24	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
26	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
28	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
32	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
34	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
36	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
38	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
40	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
42	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
44	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
46	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
48	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
50	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
52	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
54	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
56	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
58	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
60	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
62	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
64	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
66	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
68	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
70	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
72	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
74	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
76	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
78	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
80	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
82	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
84	10.0.0.10.0.0	127.0.0.1	TCP	44	5121 → 40783 [RST

```
tcp.flags.reset == 1 filter result
```

The screenshot displays the Wireshark network protocol analyzer interface. The top pane shows a list of captured packets, with packet 443 highlighted in blue. The middle pane shows the details of the selected packet, which is a TCP Segment (33319) from 194.18.156.37 to 192.168.31.49. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
2209	2.892325600	2404:6800:4000:81d::	2409:4091:7:1a13:f1b::	TCP	508	[TCP Spurious Retransmission] 443 → 28924 [PSH, ACK] Seq=382 Ack=33319
2267	2.892420500	2409:4091:7:1a13:f1b::	2404:6800:4000:1023::	TCP	1334	[TCP Retransmission] 33319 → 443 [PSH, ACK] Seq=382 Ack=33319
2350	3.015500000	2404:6800:4000:1023::	2409:4091:7:1a13:f1b::	TCP	1294	[TCP Retransmission] 443 → 33319 [ACK] Seq=1 Ack=382 Win=0 Len=0
2431	4.484531000	2404:6802:59:347::	2409:4091:7:1a13:f1b::	TCP	113	[ACK] Seq=1 Ack=382 Win=0 Len=0
2538	9.770531000	2801:4860:482b:7700::	2409:4091:7:1a13:f1b::	TCP	1294	[TCP Retransmission] 443 → 12551 [PSH, ACK] Seq=1221 Ack=33319
2639	9.770531000	2801:4860:482b:7700::	2409:4091:7:1a13:f1b::	TCP	1294	[TCP Retransmission] 443 → 12551 [ACK] Seq=1 Ack=382 Win=0 Len=0
4542	32.598110000	194.18.156.37	192.168.31.49	TLSv1.2	78	[TCP Retransmission], Application Data
4543	32.598110000	194.18.156.37	192.168.31.49	TCP	54	[TCP Retransmission] 443 → 44437 [FIN, ACK] Seq=1 Ack=382

Packet 2267: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface 0
 Ethernet II, Src: AzureWaveTec, Prio: 0x0a (2a), Dst: 08:00:00:00:00:00
 Internet Protocol Version 6, Src: 2409:4091:7:1a13:f1b::3a2b, Dst: 2404:6800:4000:1023::1023
 Transmission Control Protocol, Src Port: 33319, Dst Port: 443, Seq: 33319, Len: 0

Packet 4543: This frame is a (suspected) retransmission: Label

Packets: 4634, Displayed: 8 (0.2%)

Profile: Default

tcp.analysis.retransmission filter result